

Project F.R.I.D.A.Y.

Parts List & Cost Estimate

Voice-controlled, mobile AI assistant robot with face recognition, object identification and follow-me mode.
Working-model exhibition – 12 September 2026. Hardware lead: Shivnandan Sahu.

A. Core robot (brain, voice, face, senses)

#	Part	Purpose	Qty	Est. cost
1	ESP32 DevKit V1 (CP2102)	Robot body brain – drives the two BTS7960 motor drivers over WiFi (UDP fast path)	1	₹200
2	ESP32-S3 DevKitC-1 (N16R8)	Face & senses brain – two OLED eyes, four ultrasonics, DHT22, PIR, flame, gas, light	1	₹900
3	ESP32-CAM + programmer board	Camera / FRIDAY's eye – face recognition and object identification (thinking runs on the laptop)	1	₹1,000
4	USB microphone	Voice input to laptop	1	₹1,000
5	USB / 3.5mm speaker	Voice output	1	₹800
6	1.3" OLED display 128x64 (SH1106, I2C) - the bigger module, sold as 1.02"-1.3"	Eyes / status face (replaced the 0.96" SSD1306 pair)	2	₹700
7	HC-SR04 ultrasonic sensor	Distance – two front, two rear (obstacle avoidance, follow-me)	4	₹360
8	PIR motion sensor (HC-SR501)	Detects people approaching	1	₹150
9	DHT22 sensor	Temperature & humidity report	1	₹250
10	Flame sensor module	Fire alert – announced the instant it triggers	1	₹60
11	MQ-2 gas / smoke sensor	Gas & smoke alert	1	₹150
12	LDR light sensor module	Room brightness	1	₹40
13	5V 3A power supply / power bank	Powers ESP32 logic, the camera and the motor-driver logic side	1	₹1,200
14	LM2596 buck converter	One at 3.30 V for the S3 board, OLEDs and sensors; one at 5 V for camera and driver logic	2	₹130
15	Chassis materials (foam board / acrylic / cardboard, hot glue)	Body & head	–	₹1,000
16	Jumper wires, breadboard, headers, tape, screws	Wiring & assembly	–	₹700
17	Laptop	Main AI brain – IRIS server, speech-to-text, text-to-speech, face recognition, on-screen face	1	Already owned
18	Phone hotspot (2.4 GHz)	The network all three boards and the laptop share	1	Already owned

Subtotal - Core robot

₹8,640

B. Mobility & follow-me system

#	Part	Purpose	Qty	Est. cost
19	BO / TT gear motors with wheels	Drive wheels	4	₹500
20	BTS7960 motor driver	Controls the drive motors, one per side	2	₹700
21	Robot chassis plate (acrylic / metal)	Base for motors, body sits on top	1	₹350

#	Part	Purpose	Qty	Est. cost
22	12V battery	Higher-capacity power for motors	2	₹500
23	Battery charger	Charging the battery pack	1	₹100
24	Rocker power switch	Master on/off	1	₹40
Subtotal - Mobility & follow-me				₹2,190

C. Software & services

#	Part	Purpose	Qty	Est. cost
25	IRIS software (own code)	FastAPI server, dark hologram UI, 83 tools, command rules, memory, device link, vision	1	₹0
26	Python 3.11 + open-source libraries	FastAPI, httpx, SQLAlchemy, faster-whisper, edge-tts, openwakeword, insightface / face-recognition	-	₹0
27	Arduino IDE + ESP32 board package	Firmware for the three boards (Adafruit SSD1306, DHT, ArduinoJson libraries)	-	₹0
28	AI API - OpenRouter / Groq / Gemini / Cerebras free tiers	Cloud intelligence and the vision model for 'what is this'	-	Free tier
29	Claude Pro subscription (extra tokens)	Used to develop and debug the software	1	₹11,000
30	Hosting / VPS / domain	Not needed - everything runs on the laptop over the hotspot	-	₹0
Subtotal - Software & services				₹11,000

ESTIMATED GRAND TOTAL	₹21,830
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Hardware only: ₹10,830. Realistic range (depends on seller / brand): ₹20,700 - ₹23,600

Notes

- All prices are approximate Indian market rates (Sept 2026) from typical online electronics sellers; actual prices vary.
- Core parts already received except the microphone, speaker, ESP32-S3, flame, gas and light sensors - remaining spend on section A is roughly ₹3,000.
- Laptop, hotspot and the AI API add no cost (already owned / free tier). The Claude Pro subscription (₹11,000) is the only software cost; every other program is free and open source, and there is no server to rent. Recurring cost after the build: ₹0 per month.
- Software stack: Python on the laptop (IRIS - speech recognition, AI API, text-to-speech, face recognition, WiFi link to the boards) and Arduino C++ on the three ESP32 boards.
- Follow-me and obstacle avoidance use the four ultrasonic sensors (two front, two rear) as the primary method, with ESP32-CAM face tracking as the smart mode; the OLED eyes turn toward the person the camera finds.
- Power: motors run from the 12 V battery through the BTS7960s; the S3 board, OLEDs and every sensor run from one buck converter set to 3.30 V, so no resistors or voltage dividers are needed anywhere. All grounds are common.